

CLARA: Clip-Level Multimodal Alignment with VLM-Derived Rationales for Hateful Video Detection

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Abstract

Hateful video detection has become increasingly important with the rapid growth of video-centric social media platforms, given the serious risks that hate speech poses to both individual well-being and social cohesion. Compared with text or static multimodal content, hateful video detection remains underexplored and significantly more challenging, as hateful meaning often arises from complex interactions among multimodal cues, including speech, audio, and visual content. Moreover, such signals are often brief, implicit, and temporally dependent, making them difficult to capture using conventional video-level representations. In this work, we propose CLARA¹, a clip-level multimodal framework for hateful video detection. Instead of treating a video as a single instance, CLARA models it as a sequence of fine-grained clips, enabling more precise capture of temporally localized hateful signals. We introduce a Mixture-of-Experts clip encoder for adaptive multimodal alignment, a local-global segment contrastive objective to jointly model short-term cues and long-range temporal dependencies, and VLM-derived rationales integrated via a gated Transformer to provide high-level semantic guidance. Extensive experiments on three hateful video datasets demonstrate that CLARA consistently outperforms state-of-the-art methods. Further ablation studies and parameter analyses validate the effectiveness of each component.

Disclaimer: This paper contains sensitive content that may be disturbing to some readers.

CCS Concepts

• Computing methodologies → Artificial intelligence; Computer vision; Natural language processing.

¹The code for CLARA is available at <https://github.com/yuchenzhang-1/CLARA>.



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Figure 1: An example of hateful video. The hate speech is not fully revealed by single frame or utterance, but emerges through contextual buildup across multiple clips.

Keywords

Hateful Video Detection, Multimodal Alignment, Mixture of Experts, Contrastive Learning, Video Transformer

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1 Introduction

Hate speech poses a serious threat to both individual well-being and social cohesion. It reinforces prejudice, normalizes discrimination, and intensifies hostility toward targeted groups, thereby causing psychological harm and deepening social division [16]. With the rapid growth of social media, particularly video-centric platforms such as TikTok, hateful content in video form is becoming increasingly prevalent. Accurate and efficient hate speech detection is therefore essential for enabling timely moderation and mitigating its broader societal harms.

Hate speech detection has been widely studied in text and, more recently, in multimodal settings such as memes and image-text posts

[6, 34]. In contrast, hateful video detection remains relatively under-explored. This gap is increasingly important because videos present a substantially more challenging setting than text or static multimodal content. Hateful meaning in videos often arises from the interaction of spoken words, vocal tone, and visual scenes, with many cues being brief, context-sensitive, and deliberately indirect. Moreover, as illustrated in Figure 1, hateful intent may not be fully revealed by single frame or utterance, but instead emerge progressively through contextual buildup across multiple clips of the whole video. A segment that appears neutral in isolation may reveal clear hateful intent only when interpreted in the context of preceding and subsequent content. Such temporal dependency makes hateful video understanding fundamentally more complex and challenging.

Existing hateful video detection methods still largely follow a coarse video-level modeling paradigm, where each modality is encoded over the entire video, reduced to a global representation, fused across modalities, and then used for final classification. Although recent advances have introduced stronger multimodal learning mechanisms, including cross-modal attention [10], retrieval-augmented multimodal experts [22], and vision-language-model (VLM) reasoning [20], most existing approaches remain centered on video-level encoding and fusion. As a result, they remain limited in their ability to capture fine-grained and temporally evolving hateful signals.

In particular, three key challenges remain insufficiently addressed. First, the reliance on video-level representations prevents models from capturing fine-grained temporal units. This can be limiting, as hateful content in videos is often localized to short segments and may only become meaningful within specific contextual boundaries. Compressing the entire video into a single representation therefore risks diluting short but decisive hateful moments with long neutral content. Second, existing methods typically rely on fixed multimodal fusion strategies, imposing a single fusion rule across the entire video. However, in hateful videos, the hateful evidence may shift across modalities over time, for example from spoken language to visual context [8]. Applying a uniform fusion mechanism thus fails to adapt to such dynamic modality contributions, limiting the model’s ability to capture diverse patterns of hateful expression. Third, prior work rarely models the relationship between local cues and global context explicitly. This is a critical limitation because hateful meaning often emerges from the interaction between brief signals and broader discourse. Recent studies show that modeling both short-term and long-term temporal views can improve representation learning by preserving informative local structure while enforcing global temporal consistency [33], and similar findings have been reported in multimodal sequential learning [19]. These findings suggest that jointly modeling local and global dependencies is essential for accurately interpreting implicit and context-dependent hate.

To address these limitations, we propose **CLARA**, a **Clip-Level multimodal Alignment with VLM-derived RAtionales** for hateful video detection. Instead of treating a video as a single instance, CLARA first segments it into video clips, preserving fine-grained temporal units. It then performs clip-level multimodal alignment using a Mixture-of-Experts (MoE) encoder, enabling input-dependent fusion across modalities. To bridge local cues and global context, we introduce a local-global segment contrastive objective that encourages the model to capture both short-term signals and long-range

temporal dependencies. Finally, we incorporate VLM-derived rationales through a gated Transformer, which aggregates rationale and clip representations into a video-level representation, enabling the model to infer hateful meaning from interactions across clips under high-level semantic guidance for more effective hateful video detection.

Our contributions are summarized as follows:

- We propose CLARA, a novel clip-level multimodal framework for hateful video detection that moves beyond coarse video-level modeling by explicitly representing videos as sequences of utterance-aligned clips, allowing decisive hateful cues to be preserved and modeled more effectively.
- CLARA combines a MoE-based clip encoder for adaptive multimodal alignment, a local-global contrastive objective to capture both short-term signals and long-range temporal dependencies, and VLM-derived rationales integrated via a gated Transformer to provide high-level semantic guidance, enabling more robust hateful video detection.
- Extensive experiments on three hateful video datasets demonstrate that CLARA consistently outperforms state-of-the-art baselines, achieving superior performance in hateful video detection, while ablation studies and parameter analyses further verify the effectiveness of its key design choices.

2 Related Work

Hate speech detection was initially studied mainly in text, and later extended to multimodal settings such as memes, image-text posts, and audio-based harmful content analysis [17]. In these settings, hateful meaning is often expressed jointly across modalities rather than through a single channel. For example, multimodal meme detection methods combine textual and visual reasoning for implicit hate analysis [9, 24], while audio-based approaches explore acoustic features for hate speech recognition [18, 35]. These developments are relevant to videos, but they do not directly address the key challenge of hateful video understanding, where speech, frames, and on-screen text unfold over time, and hateful meaning often arises from their interaction.

Currently, research on hateful video detection remains relatively limited. Early studies largely treated videos as text-bearing objects, relying on transcripts or metadata such as titles, descriptions, and comments, rather than modeling videos as multimodal temporal signals [2, 21]. With the release of dedicated hateful video datasets such as HateMM [12], MultiHateClip [32], and DeHate [36], subsequent work began to incorporate transcript, audio, and visual information for video-level hateful content prediction. A representative early multimodal baseline is CAMFusion [37], which extracts textual, visual, and acoustic features using BERT [13], ViT [14], and MFCC, respectively, and fuses them through channel-wise and modality-wise operations before final classification.

More recent studies have explored stronger multimodal interaction mechanisms. For example, MM-HSD [10] enhances multimodal fusion by explicitly separating transcript and OCR text, extracting visual and audio features with ViT and Wav2Vec [4], and applying cross-modal attention before final fusion. MoRE [22] introduces modality-specific experts, augments them with retrieved hateful and non-hateful neighbors through a joint multimodal retriever, and uses

a sample-sensitive router to adapt modality contributions across short videos. HVGuard [20] instead adopts a reasoning-driven approach, where VLMs generate chain-of-thought-style rationales from video frames, transcripts, and audio emotion cues, and these rationale embeddings are then fused with modality representations through an MoE network. These studies show that stronger multimodal fusion, adaptive expert routing, retrieval augmentation, and explicit reasoning can substantially improve hateful video detection, especially for implicit and context-dependent hate.

Despite these advances, most existing methods still perform encoding and fusion at the whole video level. Each modality is compressed into a global representation before multimodal interaction and final prediction. Our work differs from this line by moving the basic modeling unit from the whole video to utterance-aligned clips, performing multimodal alignment at the clip level, and explicitly modeling the relationship between local hateful cues and broader discourse context before temporal aggregation.

3 Proposed Method

Our goal is to detect hate speech in videos by capturing fine-grained clip-level multimodal interactions across text, audio, and visual signals, enhancing local-global temporal consistency, and leveraging VLM-generated video-level rationales as high-level semantic guidance, as illustrated in Figure 2. The proposed CLARA comprises six main components: (i) utterance-aligned clip segmentation, (ii) multimodal feature extraction and a MoE-based clip encoder, (iii) local-global segment contrastive learning, (iv) VLM-driven video-level rationale generation, (v) a rationale-gated Transformer that integrates the generated rationale with clip-level embeddings to produce a video-level representation, and (vi) a video-level hate speech classifier optimized with both supervised classification loss and a self-supervised contrastive objective.

3.1 Video Clip Segmentation

Videos exhibit highly heterogeneous temporal structures, where hateful cues may be diluted when modeling at the coarse video level. To better capture fine-grained multimodal interactions, we first segment each video into a sequence of clips.

Rather than fixed length segmentation, we segment videos based on utterance boundaries, as they provide explicit semantic units that align naturally with the underlying video content. Specifically, we employ Whisper-large-v3 [28] to obtain transcriptions along with sentence-level timestamps, and use these timestamps to define clip boundaries. This ensures that each clip corresponds to a semantically coherent segment while remaining temporally aligned across text, audio, and visual modalities.

This segmentation process may introduce silent intervals between consecutive utterances. We then use the duration of each silent interval to determine whether it should be merged into an adjacent clip or retained as a standalone segment. Prior work suggests that pauses longer than 1 second are typically categorized as long pauses, and such pauses are more likely to mark boundaries between discourse units than to reflect brief within-utterance hesitations [7]. We therefore use 1 second as the threshold, merging short pauses into adjacent clips to avoid unnecessary fragmentation while retaining longer silent spans as standalone segments.

3.2 Multimodal Feature Extraction and MoE Clip Encoder

Given a video $V_j = \{C_1, C_2, \dots, C_N\}$, where N is the number of clips in video V_j , we first construct multimodal inputs for each clip $C_i \in V_j$, extract modality-specific features from text, audio, and visual streams, and then fuse them into a unified clip representation h_i^C using a MoE encoder.

Audio features. For the audio modality, we feed the audio signal of C_i , denoted as C_i^A , into the encoder of Whisper-large-v3 [28] to obtain the raw audio embedding E_i^A .

Visual features. We define a fixed frame budget S_{frame} for the entire video and allocate to each clip C_i a clip-specific budget s_i according to its duration. We then sample s_i frames $C_i^V = \{f_1^i, \dots, f_{s_i}^i\}$, and encode them using a pre-trained ViT [14] to obtain frame-level visual embedding E_i^V .

Text features. We use both transcription and OCR text as the textual input of each clip. Specifically, the transcription C_i^{trans} and the OCR text C_i^{ocr} extracted from the sampled frames C_i^V using PaddleOCR [11] are encoded using BERT [13] to obtain E_i^{trans} and E_i^{ocr} , respectively. The two embeddings are then concatenated to form the textual embedding $E_i^T = [E_i^{trans}, E_i^{ocr}]$.

MoE-based clip encoder. The extracted modality-specific embeddings are first mean-pooled and then projected into a shared latent space:

$$\tilde{E}_i^u = \phi_u(\text{MeanPool}(E_i^u)), \quad u \in \{A, V, T\}, \quad (1)$$

where $\phi_u(\cdot)$ is the corresponding modality-specific projector implemented as a two-layer MLP.

We then concatenate the modality-specific representations into a fused clip representation:

$$E_i^{Fused} = \text{Concat}(\tilde{E}_i^A, \tilde{E}_i^V, \tilde{E}_i^T). \quad (2)$$

Given the fused clip representation E_i^{Fused} , a gating network $g(\cdot)$ computes a routing distribution over M experts:

$$p_i = \text{softmax}(g(E_i^{Fused})), \quad p_i \in \mathbb{R}^M, \quad (3)$$

where $g(\cdot)$ is implemented as a linear projection layer.

Based on p_i , we select the top- K experts and compute their outputs as:

$$z_i^{(m)} = f_m(E_i^{Fused}), \quad m \in \text{TopK}(p_i), \quad (4)$$

where each expert $f_m(\cdot)$ is implemented as a two-layer MLP.

The final clip representation is obtained as a weighted combination of the selected expert outputs:

$$h_i^C = \sum_{m \in \text{TopK}(p_i)} p_i^{(m)} z_i^{(m)}. \quad (5)$$

To prevent routing collapse and encourage balanced expert utilization, we further introduce a load-balancing regularization term following prior MoE work [15, 30]. We define the importance of expert m as the average routing probability assigned to it over a batch of size B :

$$\text{Imp}(m) = \frac{1}{B} \sum_{i=1}^B p_i^{(m)}. \quad (6)$$

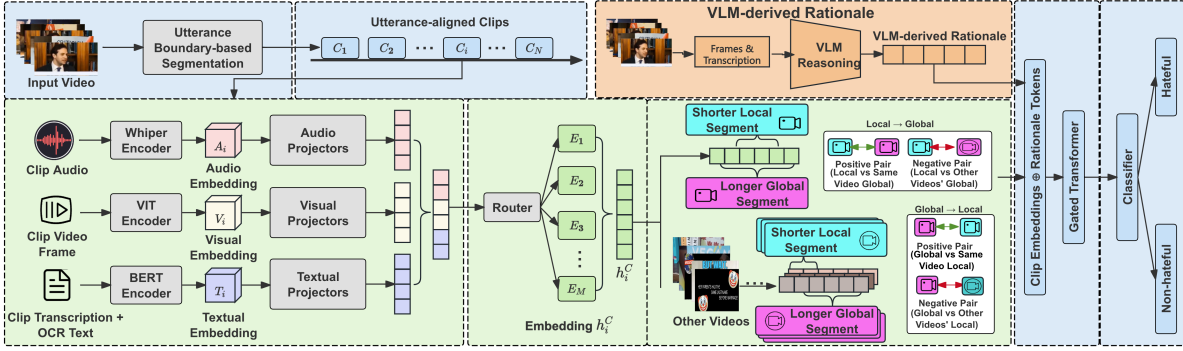


Figure 2: Overall framework of CLARA. An input video is first segmented into utterance-aligned clips, from which multimodal features are extracted and fused into clip-level representations via a MoE encoder. Local-global segment contrastive learning is then applied to enhance temporal consistency across clips. Meanwhile, video-level rationales are generated by a VLM and integrated with the clip representations through a rationale-gated Transformer to produce the final video representation, which is finally used for hateful video classification.

We further define the load of expert m as the fraction of top- K assignments routed to it:

$$\text{Load}(m) = \frac{1}{B} \sum_{i=1}^B \frac{1}{K} \mathbb{I}[m \in \text{TopK}(p_i)], \quad (7)$$

where $\mathbb{I}[\cdot]$ is the indicator function, which equals 1 if the condition holds and 0 otherwise. The load-balancing loss is then given by

$$\mathcal{L}_{\text{LB}} = M \sum_{m=1}^M \text{Imp}(m) \text{Load}(m). \quad (8)$$

3.3 Local-Global Segment Contrastive Learning

We introduce a local-global segment contrastive learning objective to encourage local segments to align with their corresponding global context while remaining distinguishable from segments of other videos, by enforcing consistency between short local segments and longer global segments within the same video and pushing apart segments from different videos.

Given a video $V_j = \{C_1, \dots, C_N\}$ with corresponding clip embeddings $\{h_1^C, \dots, h_N^C\}$, we construct two types of segment-level representations: (i) local segments consisting of consecutive short clips, and (ii) global segments covering a longer temporal span. Their lengths are determined as $s_\ell = \max(1, \lfloor r_\ell N \rfloor)$ and $s_g = \max(1, \lfloor r_g N \rfloor)$, where r_ℓ and r_g are the local and global ratios and $r_g > r_\ell$.

Specifically, we sample the local segment and the global segment separately from the same video, using two independently selected valid start indices rather than forcing them to share the same temporal boundary or requiring one to contain the other. The resulting segments are $\mathcal{S}_\ell = \{h_{i_\ell}^C, \dots, h_{i_\ell+s_\ell-1}^C\}$ and $\mathcal{S}_g = \{h_{i_g}^C, \dots, h_{i_g+s_g-1}^C\}$, where i_ℓ and i_g are independently drawn valid start indices.

We aggregate each segment into a segment-level representation via mean pooling:

$$h_\ell = \text{MeanPool}(\mathcal{S}_\ell), \quad h_g = \text{MeanPool}(\mathcal{S}_g). \quad (9)$$

We treat the local and global segment representations from the same video as a positive pair, and use segment representations from other videos in the batch as negatives. Specifically, we adopt a

bidirectional InfoNCE objective:

$$\mathcal{L}_{\text{CL}} = \frac{1}{2} (\mathcal{L}_{\ell \rightarrow g} + \mathcal{L}_{g \rightarrow \ell}), \quad (10)$$

where

$$\mathcal{L}_{\ell \rightarrow g} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(\text{sim}(h_\ell^{(i)}, h_g^{(i)})/\tau)}{\sum_{j=1}^B \exp(\text{sim}(h_\ell^{(i)}, h_g^{(j)})/\tau)}, \quad (11)$$

and

$$\mathcal{L}_{g \rightarrow \ell} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(\text{sim}(h_g^{(i)}, h_\ell^{(i)})/\tau)}{\sum_{j=1}^B \exp(\text{sim}(h_g^{(i)}, h_\ell^{(j)})/\tau)}. \quad (12)$$

Here, $\text{sim}(\cdot, \cdot)$ denotes cosine similarity and τ is a temperature parameter.

3.4 Gated Transformer for Video Encoding

To model inter-clip temporal dependencies at the video level, we further introduce a rationale-gated Transformer encoder. This module incorporates video-level rationales generated by VLMs to guide the contextualization and aggregation of clip representations

Video-level VLM rationale. For each video V_j , we generate a video-level rationale using Qwen3VL-8B-Instruct [5] conditioned on both visual and textual evidence. To obtain stable and interpretable outputs, we adopt a two-step prompting strategy.

First, we ask the VLM to perform an objective analysis of the video: (i) describe what is visible in the sampled frames, (ii) summarize the key messages conveyed by the provided text, (iii) provide an overall objective summary by combining the visual and textual evidence, and (iv) explain how the visuals and the text relate to each other. The inputs to this step include the video title, the transcription, and 20 sampled frames. The resulting output is a structured objective analysis, denoted as OA_j .

Then, we feed the same inputs as in the previous step, together with the objective analysis OA_j , into the VLM and ask it to perform hateful content verification: (i) determine whether the video is hateful or non-hateful, (ii) identify whether the hatefulness is explicit or implicit if hate is detected, and (iii) provide the main reasons supporting the decision, focusing on observable or stated evidence

rather than speculation. Together, these two steps produce the final video-level rationale, denoted as RA_j , which is used to obtain the rationale representation for subsequent video-level encoding.

Rationale-Gated Transformer. Given a video $V_j = \{C_1, \dots, C_N\}$ with corresponding clip representations $\{h_1^C, \dots, h_N^C\}$ and the video-level rationale RA_j , we model temporal dependencies using a rationale-gated Transformer encoder. Our design is motivated by the observation that standard Transformers perform fine-grained token interactions via self-attention, but do not explicitly regulate the relative contribution of different information sources before such interactions [31]. Inspired by feature-wise conditioning and conditional information modulation [1, 23, 27], we introduce a lightweight source-level gating mechanism to apply coarse-grained modulation to the rationale and clip representations prior to Transformer encoding. Such coarse-to-fine control is particularly beneficial for hateful video detection, where the relative importance of clip level evidence and high level semantic reasoning can vary substantially across videos.

Specifically, for a video V_j , we first obtain its rationale representation H_j^{RA} by encoding its rationale RA_j using BERT to get its embedding E_j^{RA} , and then projecting it into the same hidden space as the clip representations:

$$H_j^{RA} = \text{Linear}(E_j^{RA}). \quad (13)$$

Let $H_j^C = \{h_1^C, h_2^C, \dots, h_N^C\}$ denote the sequence of clip representations for video V_j . We then compute two source-level gates, one for the rationale branch and the other for the clip branch, based on their pooled representations:

$$g_j^{RA} = \sigma(\text{Linear}(\text{MeanPool}(H_j^{RA}))), \quad (14)$$

$$g_j^C = \sigma(\text{Linear}(\text{MeanPool}(H_j^C))), \quad (15)$$

where $\sigma(\cdot)$ denotes the sigmoid function.

The two gates are used to modulate the rationale and clip representations before temporal encoding:

$$\hat{H}_j^{RA} = g_j^{RA} H_j^{RA}, \quad \hat{h}_i^C = g_j^C h_i^C \quad (h_i^C \in H_j^C). \quad (16)$$

We then concatenate the gated rationale and clip representations, and feed them into a Transformer encoder with positional encoding to obtain the final video representation of V_j :

$$H_j^{\text{final}} = \text{Transformer}(\text{PosEnc}([\hat{H}_j^{RA}, \hat{h}_1^C, \dots, \hat{h}_N^C])). \quad (17)$$

3.5 Training Objective

Given the final video representation H_j^{final} , we perform video-level classification using an MLP classifier:

$$\hat{y}_j = \text{softmax}(\text{MLP}(H_j^{\text{final}})), \quad (18)$$

where the MLP consists of two linear layers with a ReLU activation and dropout in between.

We optimize the model using a standard cross-entropy loss for supervised hateful video classification:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{B} \sum_{j=1}^B y_j \log \hat{y}_j, \quad (19)$$

where y_j is the ground-truth label and B is the batch size.

In addition, we have $\mathcal{L}_{\text{LB}}$ from MoE clip encoder and $\mathcal{L}_{\text{CL}}$ from local-global segment contrastive learning. The overall training objective is:

$$\mathcal{L} = (1 - \lambda_{\text{CL}})\mathcal{L}_{\text{CE}} + \lambda_{\text{CL}}\mathcal{L}_{\text{CL}} + \lambda_{\text{LB}}\mathcal{L}_{\text{LB}}, \quad (20)$$

where λ_{CL} and λ_{LB} control the contributions of the contrastive and load-balancing terms, respectively.

4 Experiments

4.1 Experimental Setup

Table 1: Dataset Statistics.

	HateMM	MHC_CN	MHC_EN	DeHate
Hateful videos	427	298	296	2,120
Non-hateful Videos	638	608	607	4,568
Total videos	1,065	906	903	6,688
Platforms	BitChute	Bilibili	YouTube	BitChute, TikTok

Datasets. We conduct experiments on three publicly available hateful video datasets, namely HateMM [12], MultiHateClip [32], and DeHate [36], as shown in Table 1. For MultiHateClip, we follow its language split into the Chinese subset (MHC_CN) and the English subset (MHC_EN) and adopt the binary classification setting from the original paper by grouping hate and offensive samples into one class and treating non-hate samples as the other class. These datasets differ in scale, language, content, and source platforms, covering BitChute, YouTube, Bilibili, and TikTok, and therefore provide a diverse benchmark for evaluating hateful video detection.

Baselines. We compare CLARA against a comprehensive set of baselines spanning both unimodal and multimodal settings. For unimodal baselines, we include strong modality-specific encoders: BERT [13] and RoBERTa [26] for text, ViViT [3] and ViT [14] for video, and Wav2Vec [4] and Whisper [29] for audio. For multimodal baselines, we consider both general-purpose VLMs and task-specific hateful video detection methods. We evaluate LLaVA1.5-7B [25] and Qwen3VL-8B-Instruct [5] through direct prompting, using the same prompting procedure and input information as in rationale generation, and take the final VLM decision as the hateful or non-hateful prediction. For task-specific baselines, we include HateMM* [12], MHC* [32], and DeHate* [36], which are the original benchmark baselines introduced with their respective datasets. These methods follow a similar pipeline in which modality-specific features are extracted separately, and then concatenated into a unified video representation for classification. We further include more recent multimodal methods. MoRE [22] incorporates retrieval-augmented contextual knowledge with modality-specific experts and sample-sensitive multimodal fusion. HVGuard [20] introduces VLM-based Chain-of-Thought reasoning for hate interpretation and uses a MoE network for multimodal fusion. MM-HSD [10] integrates video, audio, transcripts, and OCR text, and applies cross-modal attention to model inter-modal interactions.

Implementation Details. For all baselines, unless otherwise specified, we follow the parameter settings reported in their original papers. For unimodal baselines, features extracted by each encoder are fed into the same classifier as in CLARA. For HVGuard, we

Table 2: Overall Performance in terms of Acc, M-F1, M-Pre and M-Rec. Best in bold and second-best is underlined.

Method	HateMM		MHC_CN		MHC_EN		DeHate	
	Acc	M-F1	Acc	M-F1	Acc	M-F1	Acc	M-F1
BERT	0.724 ± 0.028	0.709 ± 0.032	0.696 ± 0.006	0.635 ± 0.019	0.674 ± 0.010	0.417 ± 0.020	0.651 ± 0.006	0.522 ± 0.030
Roberta	0.729 ± 0.014	0.718 ± 0.025	0.671 ± 0.003	0.402 ± 0.001	0.672 ± 0.002	0.402 ± 0.001	0.683 ± 0.000	0.406 ± 0.000
MFCC	0.653 ± 0.048	0.598 ± 0.066	0.646 ± 0.014	0.511 ± 0.020	0.656 ± 0.018	0.463 ± 0.031	0.677 ± 0.003	0.441 ± 0.030
Wav2vec	0.696 ± 0.013	0.678 ± 0.016	0.675 ± 0.003	0.403 ± 0.001	0.674 ± 0.008	0.403 ± 0.003	0.683 ± 0.001	0.407 ± 0.001
ViT	0.700 ± 0.031	0.699 ± 0.031	0.701 ± 0.007	0.553 ± 0.024	0.678 ± 0.014	0.438 ± 0.026	0.685 ± 0.010	0.542 ± 0.018
ViViT	0.718 ± 0.012	0.702 ± 0.020	0.673 ± 0.032	0.558 ± 0.030	0.688 ± 0.019	0.546 ± 0.037	0.687 ± 0.004	0.542 ± 0.022
LLaVA	0.688 ± 0.013	0.439 ± 0.015	0.669 ± 0.003	0.401 ± 0.001	0.673 ± 0.003	0.402 ± 0.001	0.687 ± 0.002	0.415 ± 0.007
Qwen3VL	0.738 ± 0.035	0.727 ± 0.035	0.709 ± 0.041	0.597 ± 0.074	0.708 ± 0.014	0.588 ± 0.027	0.657 ± 0.013	0.595 ± 0.012
HateMM*	0.771 ± 0.033	0.770 ± 0.033	0.739 ± 0.035	0.668 ± 0.046	0.678 ± 0.022	0.542 ± 0.030	0.695 ± 0.021	0.613 ± 0.016
MHC*	0.793 ± 0.023	0.782 ± 0.023	0.738 ± 0.020	0.669 ± 0.030	0.688 ± 0.031	0.600 ± 0.047	0.688 ± 0.017	0.622 ± 0.016
DeHate*	0.754 ± 0.037	0.736 ± 0.044	0.688 ± 0.009	0.595 ± 0.022	0.675 ± 0.034	0.554 ± 0.053	0.679 ± 0.015	0.590 ± 0.006
MoRE	0.813 ± 0.030	0.803 ± 0.032	0.691 ± 0.018	0.539 ± 0.108	0.697 ± 0.017	0.584 ± 0.070	<u>0.711 ± 0.011</u>	0.616 ± 0.027
HVGuard	0.815 ± 0.026	0.813 ± 0.027	<u>0.754 ± 0.024</u>	<u>0.707 ± 0.046</u>	<u>0.710 ± 0.030</u>	0.632 ± 0.034	0.707 ± 0.003	0.586 ± 0.034
MM-HSD	<u>0.839 ± 0.028</u>	<u>0.833 ± 0.029</u>	0.683 ± 0.016	0.654 ± 0.019	0.688 ± 0.017	<u>0.653 ± 0.017</u>	0.689 ± 0.018	<u>0.626 ± 0.015</u>
CLARA	0.879 ± 0.022	0.872 ± 0.024	0.779 ± 0.016	0.734 ± 0.026	0.763 ± 0.035	0.727 ± 0.042	0.735 ± 0.009	0.659 ± 0.012
Method	M-Pre	M-Rec	M-Pre	M-Rec	M-Pre	M-Rec	M-Pre	M-Rec
BERT	0.719 ± 0.030	0.705 ± 0.033	0.697 ± 0.014	0.629 ± 0.016	0.475 ± 0.196	0.503 ± 0.004	0.617 ± 0.012	0.543 ± 0.019
Roberta	0.722 ± 0.024	0.703 ± 0.027	0.336 ± 0.001	0.500 ± 0.000	0.336 ± 0.001	0.500 ± 0.000	0.342 ± 0.000	0.500 ± 0.000
MFCC	0.653 ± 0.085	0.608 ± 0.052	0.550 ± 0.027	0.528 ± 0.015	0.536 ± 0.046	0.512 ± 0.012	0.527 ± 0.036	0.508 ± 0.010
Wav2vec	0.688 ± 0.016	0.676 ± 0.017	0.337 ± 0.002	0.500 ± 0.000	0.338 ± 0.003	0.498 ± 0.003	0.542 ± 0.245	0.500 ± 0.001
ViT	0.701 ± 0.033	0.699 ± 0.031	0.669 ± 0.011	0.570 ± 0.014	0.644 ± 0.195	0.512 ± 0.010	0.610 ± 0.023	0.555 ± 0.012
ViViT	0.711 ± 0.011	0.703 ± 0.022	0.619 ± 0.054	0.566 ± 0.024	0.634 ± 0.044	0.563 ± 0.026	0.615 ± 0.009	0.556 ± 0.013
LLaVA	0.678 ± 0.120	0.513 ± 0.009	0.335 ± 0.001	0.498 ± 0.002	0.337 ± 0.001	0.500 ± 0.000	0.527 ± 0.091	0.501 ± 0.003
Qwen3VL	0.726 ± 0.035	0.736 ± 0.036	0.705 ± 0.088	0.605 ± 0.057	<u>0.729 ± 0.029</u>	0.589 ± 0.020	0.598 ± 0.013	0.593 ± 0.011
HateMM*	0.772 ± 0.035	0.771 ± 0.033	0.706 ± 0.051	0.659 ± 0.042	0.624 ± 0.053	0.557 ± 0.016	0.639 ± 0.029	0.608 ± 0.015
MHC*	0.788 ± 0.025	0.779 ± 0.021	0.704 ± 0.026	0.660 ± 0.027	0.630 ± 0.043	0.600 ± 0.038	0.631 ± 0.020	0.618 ± 0.015
DeHate*	0.748 ± 0.039	0.733 ± 0.045	0.635 ± 0.013	0.595 ± 0.017	0.605 ± 0.069	0.563 ± 0.043	0.614 ± 0.014	0.588 ± 0.006
MoRE	0.806 ± 0.032	0.801 ± 0.034	0.588 ± 0.146	0.572 ± 0.065	0.653 ± 0.027	0.595 ± 0.052	<u>0.668 ± 0.027</u>	0.614 ± 0.023
HVGuard	0.811 ± 0.026	0.814 ± 0.028	<u>0.725 ± 0.031</u>	<u>0.707 ± 0.056</u>	0.672 ± 0.062	0.628 ± 0.032	0.657 ± 0.006	0.590 ± 0.026
MM-HSD	<u>0.835 ± 0.030</u>	<u>0.833 ± 0.029</u>	0.652 ± 0.019	0.662 ± 0.023	0.652 ± 0.017	<u>0.658 ± 0.019</u>	0.654 ± 0.013	<u>0.626 ± 0.016</u>
CLARA	0.874 ± 0.022	0.872 ± 0.028	0.760 ± 0.030	0.726 ± 0.032	0.732 ± 0.041	0.725 ± 0.043	0.697 ± 0.016	0.650 ± 0.011

implement the VLM-based reasoning using Qwen3VL instead of GPT-4o for fair comparison.

For CLARA, the total number of frames per video, S_{frame} , is set to 40. We use BERT-base Chinese² for Chinese text, and BERT-base uncased³ for English text and VLM-derived rationales. The clip encoder adopts an MoE architecture with 8 experts and top-3 routing, and the weight of the load-balancing loss, λ_{LB} , is set to 0.01. For local-global contrastive learning, the local and global segment ratios, r_l and r_g , are set to 0.2 and 0.8, respectively, and the contrastive loss weight λ_{CL} is set to 0.3. CLARA is trained with AdamW using a learning rate of $3e-5$, weight decay of $1e-4$, and a warmup ratio of 0.05, with a batch size of 32.

All experiments are conducted under 5-fold cross-validation with a train: validation: test split of 7:1:2. All models are trained up to 50 epochs with early stopping on two NVIDIA L40 GPUs.

Evaluation Metrics. We evaluate each model’s performance with regard to accuracy (Acc), macro-precision (M-Pre), macro-recall (M-Rec) and macro-F1 (M-F1).

²<https://huggingface.co/google-bert/bert-base-chinese>

³<https://huggingface.co/google-bert/bert-base-uncased>

4.2 Overall Results

Table 2 presents the overall performance of all methods on all datasets. CLARA delivers the strongest performance consistently across all datasets and all four evaluation metrics, highlighting the effectiveness and robustness of our approach for hateful video detection.

Several clear trends can be observed. First, unimodal methods are consistently weaker than multimodal ones, indicating that hateful video detection cannot be reliably solved from a single modality alone. Second, VLMs such as LLaVA and Qwen3VL are stronger than most unimodal baselines, but still fall short of task-specific methods, indicating that hateful video detection requires more specialized modeling of fine-grained multimodal features. Third, among multimodal hateful video detection methods, MM-HSD, MoRE, and HVGuard are the strongest baselines, but CLARA consistently outperforms them across all datasets. Compared with the second-best method, CLARA achieves absolute improvements of 2.4% to 5.3% in Acc, 2.7% to 7.4% in M-F1, 0.3% to 3.9% in M-Pre, and 2.2% to 9.7% in M-Rec across all datasets. These consistent gains establish CLARA as a new state-of-the-art method for hateful video detection.

The superior performance of CLARA suggests that effective hateful video detection depends not only on leveraging multiple modalities, but also on when and at what temporal granularity they are aligned. Although methods such as HVGuard and MM-HSD also leverage adaptive fusion and reasoning-enhanced multimodal modeling, CLARA stands out by shifting multimodal interaction from the whole-video level to the utterance-aligned clip level. This design mitigates a key limitation of global video modeling, where short but decisive hateful cues can be diluted by long neutral segments. By aligning modalities before global compression, CLARA allows the model to focus on the clips where hateful meaning actually emerges.

Moreover, CLARA explicitly links local evidence with broader discourse structure. The local-global contrastive objective encourages short clip sequences to stay semantically consistent with longer contextual segments from the same video, while the gated Transformer models dependencies across clips as hateful meaning unfolds over time. The rationale signal further guides interpretation at the sequence level, rather than being introduced only after the video has already been compressed into a single representation. In summary, these design choices enable CLARA to more effectively align and contextualize fine-grained multimodal evidence, which is critical for hateful video detection.

4.3 Ablation Study

Table 3: Ablation Results in terms of Accuracy and M-F1.

Method	HateMM		MHC_CN		MHC_EN		DeHate	
	Acc	M-F1	Acc	M-F1	Acc	M-F1	Acc	M-F1
w/o A	0.751	0.735	0.704	0.663	0.701	0.667	0.662	0.557
w/o V	0.743	0.739	0.705	0.672	0.688	0.581	0.651	0.510
w/o T	0.712	0.707	0.690	0.653	0.669	0.468	0.668	0.532
w/o MoE	0.789	0.780	0.738	0.679	0.735	0.669	0.708	0.637
w/o CL	0.821	0.802	0.757	0.707	0.749	0.679	0.716	0.641
w/o Rationale	0.815	0.803	0.708	0.645	0.727	0.719	0.683	0.506
w/o Transformer	0.794	0.788	0.672	0.502	0.674	0.633	0.713	0.622
CLARA	0.879	0.872	0.779	0.734	0.763	0.727	0.735	0.659

To evaluate the contribution of each component in CLARA, we perform an ablation study by removing one component at a time from the full model. The variants ‘w/o A’, ‘w/o V’, and ‘w/o T’ remove the audio, visual, and textual modalities, respectively. ‘w/o MoE’ replaces the proposed MoE clip encoder with direct concatenation followed by an MLP projection. ‘w/o CL’ removes the local-global contrastive objective. ‘w/o Rationale’ excludes the VLM-derived rationale. ‘w/o Transformer’ replaces the gated Transformer with mean pooling over clip representations. The results are reported in Table 3.

The results show that removing any component consistently reduces performance, indicating that the full model benefits from the interaction of all design choices. Dropping any modality leads to clear degradation across datasets, confirming that hateful video detection depends on complementary multimodal evidence rather than any single source alone. Among the modality variants, removing text causes the largest overall decline, while removing visual information also hurts performance noticeably. Similar trends are observed for the architectural components. Replacing the MoE with concatenation and MLP fusion weakens performance on all datasets, showing the value of adaptive clip-level fusion. Removing contrastive learning leads to consistent but smaller drops, suggesting that local-global

alignment provides useful regularization beyond the base architecture. Removing rationale guidance further degrades performance, especially on M-F1, indicating that global semantic guidance helps improve balanced prediction. The performance drop becomes even more evident when the gated Transformer is replaced by mean pooling, particularly on MHC_CN, which highlights the importance of explicit temporal modeling for capturing how hateful meaning unfolds across clips. Overall, the best results are achieved only when all components are included, demonstrating that multimodality features, clip-level alignment, contrastive learning, rationale guidance, and temporal aggregation contribute complementary benefits to hateful video detection.

4.4 Parameter Analysis

MoE configuration. We examine the impact of the MoE routing configuration by varying the number of experts in {3, 5, 8, 10, 15, 20} and the top- k value in {1, 2, 3}. Figure 3a visualizes the resulting M-F1 scores as a heatmap, and warmer colors indicate better performance. The result reveals several clear patterns. First, configurations with a moderate number of experts, particularly around 8 to 10, consistently produce stronger results than very small or very large expert pools, with the peak performance achieved at 10 experts and top-2. This suggests that too few experts may limit the model ability to capture diverse multimodal patterns, whereas too many experts do not yield further gains and may instead weaken effective utilization. Second, the routing sparsity controlled by top- k also plays an important role. Using top-1 tends to underperform in many cases, indicating that relying on only one expert can overly restrict collaboration. In contrast, larger routing choices such as top-3 do not bring further improvement over top-2, suggesting that activating too many experts may reduce specialization. Overall, the results show that a balanced MoE configuration, which provides sufficient expert diversity while preserving sparse and selective routing, offers the best trade-off for CLARA.

Local-global ratio. We further examine the impact of the local and global segment ratios used in contrastive learning, where the local ratio r_ℓ is varied over {0.1, 0.2, 0.3, 0.4} and the global ratio r_g over {0.6, 0.7, 0.8, 0.9}, as shown in Figure 3b. The results do not show a monotonic trend with respect to either ratio. Instead, performance is sensitive to their combination, with the strongest performance concentrated around $r_\ell = 0.2$, while $r_\ell = 0.4$ leads to consistently weaker results under most global settings. This pattern suggests that the local segment should remain sufficiently short to preserve fine-grained temporal cues. When the local ratio becomes too large, the distinction between local and global views becomes less clear, which weakens the contrastive signal. On the global side, increasing r_g is generally beneficial, but only when paired with an appropriate local ratio. Overall, these results support the design choice of using distinct local and global temporal views, where a clear separation between the two is important for learning effective temporal representations.

Contrastive learning weight. We analyze the influence of the contrastive loss weight λ_{CL} by varying it from 0.1 to 0.9, as shown in Figure 3c. The result shows that the performance improves as λ increases from small values, reaches its best range at a moderate

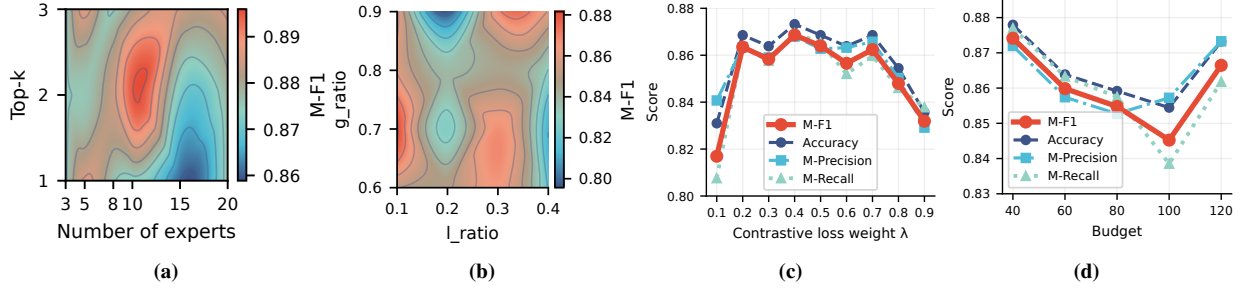


Figure 3: Parameter study of CLARA on HateMM. (a) the effect of the MoE configuration, including the number of experts and top-k routing, (b) the effect of the local and global temporal ratios used in contrastive learning, (c) the effect of the contrastive learning weight λ_{CL} , and (d) the effect of the training budget.

setting, and then gradually declines when λ_{CL} becomes too large. In particular, the best overall result is obtained around $\lambda_{CL} = 0.4$. This trend indicates that CLARA benefits most from a balanced trade-off between the contrastive objective and the main classification objective. When λ_{CL} is too small, such as 0.1, the contribution of contrastive learning is limited, preventing the model from fully exploiting temporal alignment. In contrast, when λ_{CL} becomes too large, especially beyond 0.7, optimization places excessive emphasis on alignment learning at the expense of classification, which leads to degraded performance.

Total Frame. Finally, we evaluate CLARA under different total frame budgets ranging from 40 to 120. As illustrated in Figure 3d, the performance does not improve monotonically with increasing budget. CLARA achieves the best M-F1 when the budget is set to 40, while larger budgets does not necessarily lead to better performance. This observation suggests that, at least on the HateMM dataset, a relatively small frame budget is sufficient for CLARA to capture the key multimodal cues for hateful video detection. Increasing the budget further does not provide additional benefits and may instead introduce redundant or less informative signals, which can negatively affect representation learning.

4.5 MoE Routing Analysis

To better understand how the MoE module allocates computation across experts, we compare two routing statistics on the test set of HateMM dataset, namely *Gate* and *Top-k*. *Gate* denotes the raw routing probabilities produced by the gating network before sparsification, reflecting the router’s initial preference over all experts. In contrast, *Top-k* denotes the effective routing probabilities after retaining only the top-k experts for each clip and renormalizing their scores, thus representing the actual expert usage during inference. Figure 4 visualizes the weighted expert ratio under these two distributions, where each bar indicates the proportion of total routing mass assigned to a given expert across all test clips.

As shown in Figure 4a, the gate distribution is relatively balanced across experts, with most experts receiving a similar proportion of the total gating mass. This suggests that, before sparsification, the router does not exhibit an extremely peaked preference and still assigns non-negligible probability to multiple experts. However, after top-k selection, the effective routing distribution becomes much

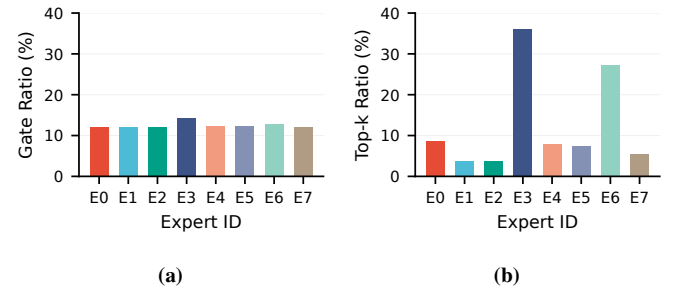


Figure 4: Expert routing statistics of the MoE. : (a) the distribution measured by gate probability ratio. (b) the distribution of experts measured by top-k selection ratio.

more concentrated. In particular, as shown in Figure 4b, Expert 3 and Expert 6 receive substantially larger routing mass than the others, while the remaining experts are markedly suppressed. This pattern indicates that the top-k routing operation amplifies relatively small differences in the original gate scores, converting a moderately balanced soft distribution into a sparse and selective routing pattern.

Overall, these results suggest that CLARA does not rely on uniform expert usage. Instead, its MoE router first maintains a soft preference over multiple experts and then, through top-k sparsification, focuses computation on a small subset of experts that are consistently preferred. This behavior is consistent with the intended design of MoE, where sparse routing encourages selective expert specialization and more targeted processing at the clip level.

5 Conclusion

In this paper, we present **CLARA**, a clip-level multimodal framework for hateful video detection. Instead of compressing an entire video into a single representation, CLARA models fine-grained clips and their multimodal interactions over time. It integrates a MoE-based clip encoder for adaptive multimodal alignment, a local-global contrastive objective for cross-scale temporal consistency, and VLM-derived rationales through a gated Transformer for semantic guidance. Experiments on three hateful video datasets demonstrate consistent improvements over state-of-the-art methods, while ablation and sensitivity studies further verify the effectiveness of the proposed components.

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